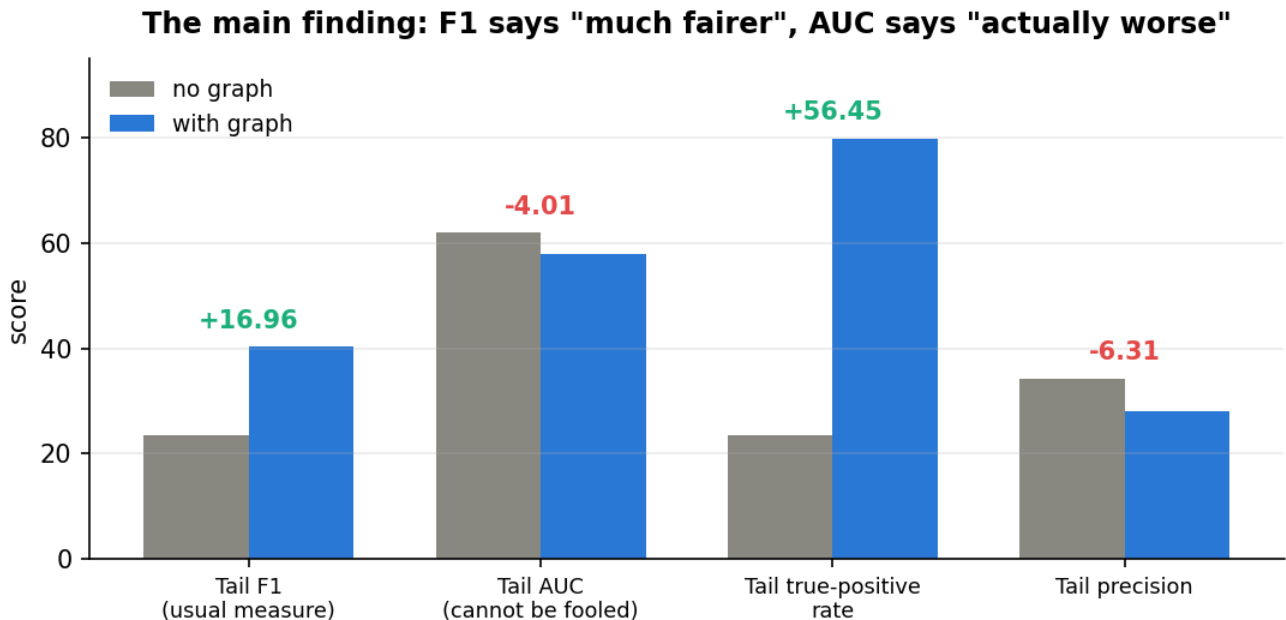


Are Fairness Scores in Crime Prediction Telling Us the Truth?

The finding in one picture



Chicago. A graph model compared against an identical model with no graph. 5 runs each.

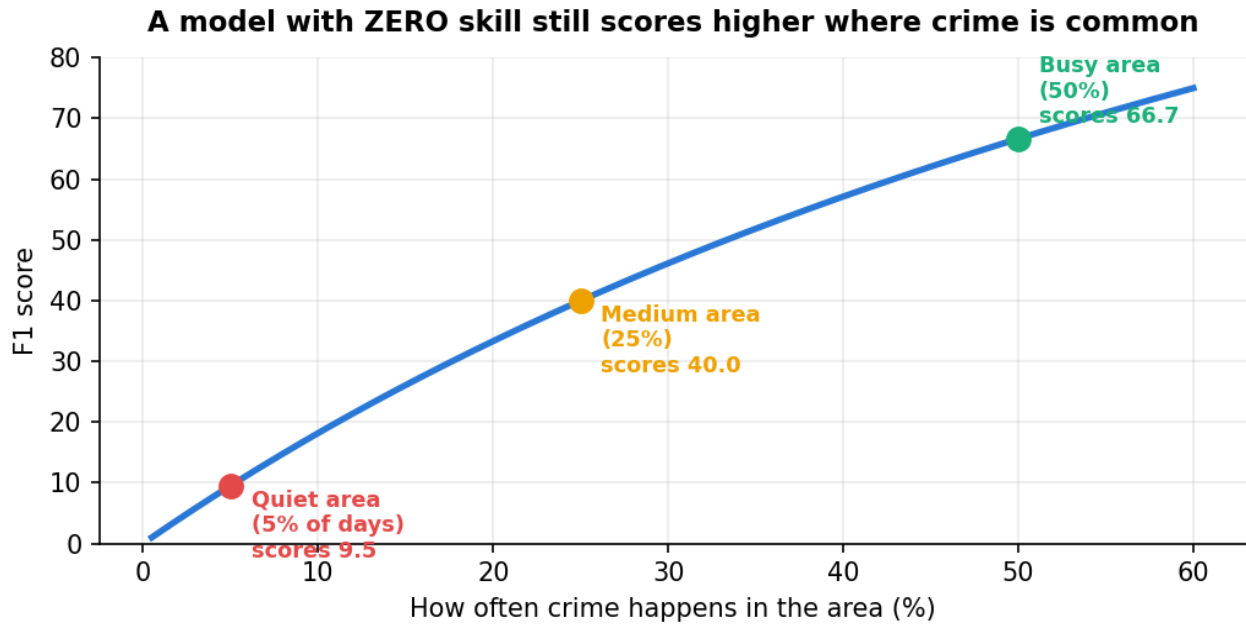
We changed our model and the usual fairness score for poor neighbourhoods jumped by **+16.96 points**. It looked like we had fixed the fairness problem.

We checked with AUC, a score that **cannot be fooled** by how rare crime is. AUC went **down 4.01 points**. The model got *worse*.

The third bar explains it. The model went from saying "crime here" 23% of the time to **80% of the time** in poor neighbourhoods. It was not finding more crime. It was guessing more, and the usual score rewarded that.

If this model were deployed, it would flag 8 out of 10 poor neighbourhoods as high-crime, while being less accurate than before — and the fairness dashboard would show a big improvement.

Why this happens



The F1 score of a model that has no skill at all and simply says "crime" every day.

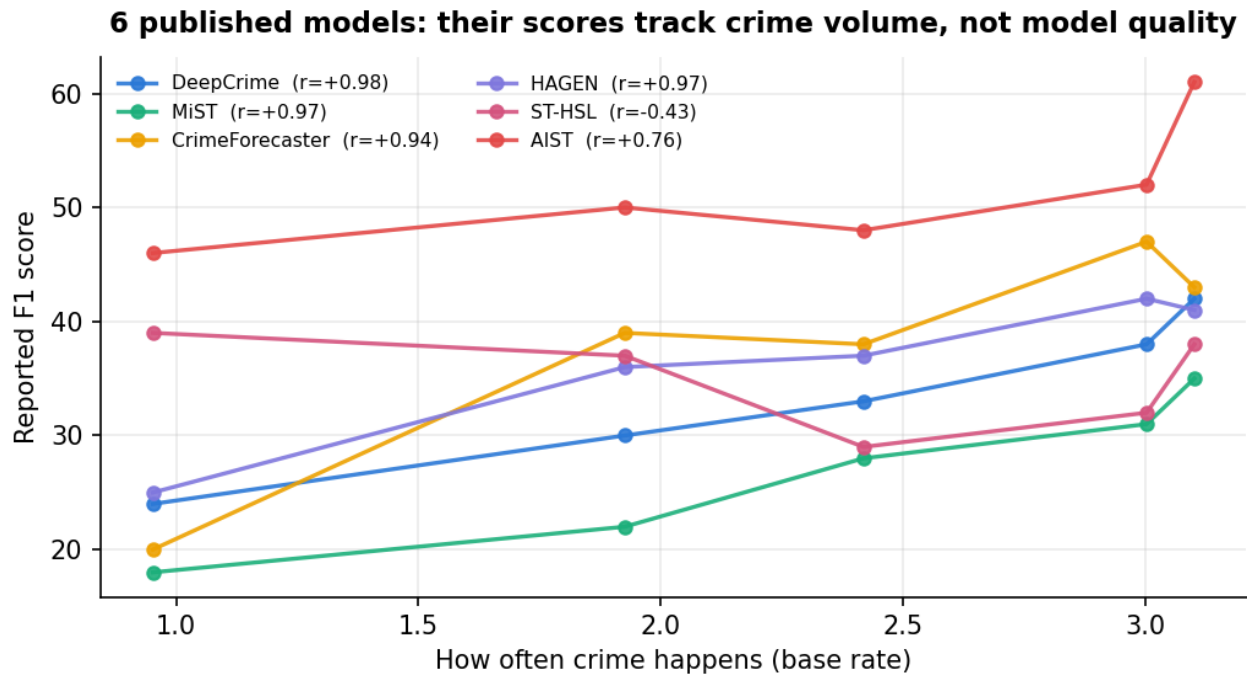
The F1 score gives **free points** where crime is common. A model that does nothing at all scores 66.7 in a busy area and only 9.5 in a quiet one — a 57-point "fairness gap" from a model that never looks at the data.

So when a paper reports "69 in busy areas, 28 in quiet areas", most of that gap was going to appear no matter how good the model was.

This part is not our discovery

The formula is known — Davis & Goadrich (2006), Boyd et al. (2012), Chouldechova (2017). We cite it as background. **Our contribution is showing that this is happening throughout crime-prediction research, and that the obvious fix does not work.**

Evidence 1 — it explains other researchers' published results



Six published crime-prediction models. Each line is one model across five area sizes.

Every line slopes upward. These models score well where crime is common and badly where it is rare — which is what a *skill-free* model would also do. Across the papers, reported scores track crime volume at $r = 0.899$. For the FedCrime paper specifically, $r = 0.998$.

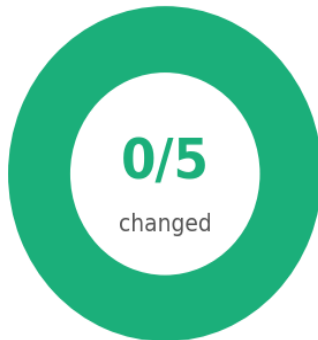
ST-HSL is the exception ($r = -0.43$), which is useful: it shows the test can tell the difference, rather than flagging everything.

Where we checked	Fake gap	Link to sparsity	What it means
Los Angeles	42.8		Gap appears in every city
Chicago	45.9	$r = +0.835$	The sparser the data,
New York	19.9	across cities	the bigger the fake gap
San Francisco	48.8		
Chicago 311 (not crime)	same pattern	—	Not a crime-data quirk

Evidence 2 — it changes real decisions

Does the flawed metric change real decisions?

Choosing the BEST MODEL



no decisions changed

Choosing WHICH AREA needs help



every decision changed

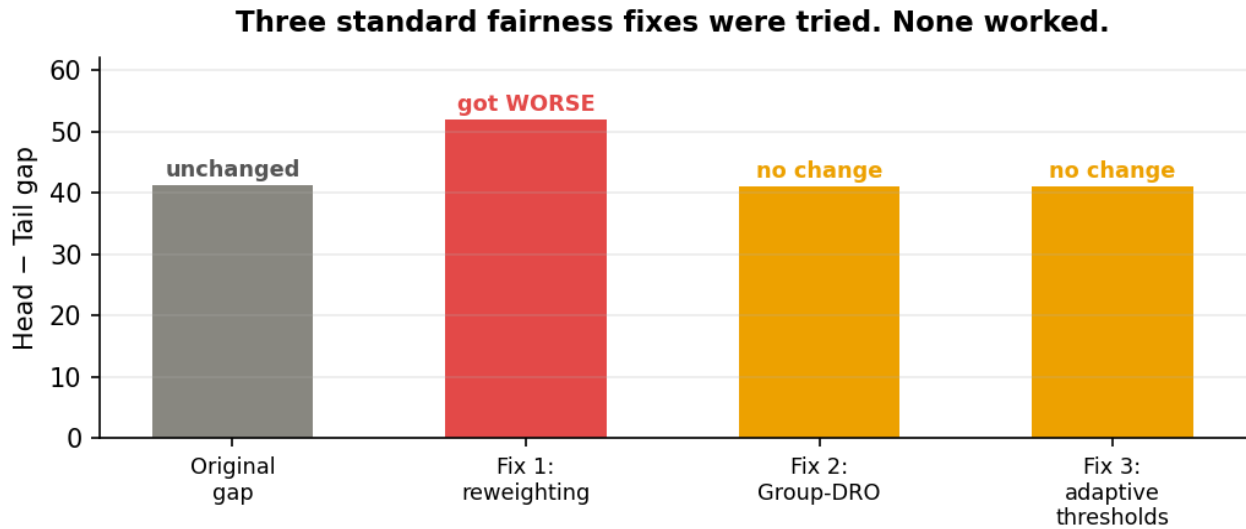
Tested on six published models.

Good news: picking which method is best is unaffected. Within one area group all models get the same free points, so the ranking survives.

Bad news: deciding *which neighbourhood needs help* flipped for every single model. Under the usual score the poorest areas always look worst — almost by construction — so extra resources get sent there whether or not the model is actually failing them.

The flaw does not mislead you about which method to use. It misleads you about where the model is failing — which is the whole fairness question.

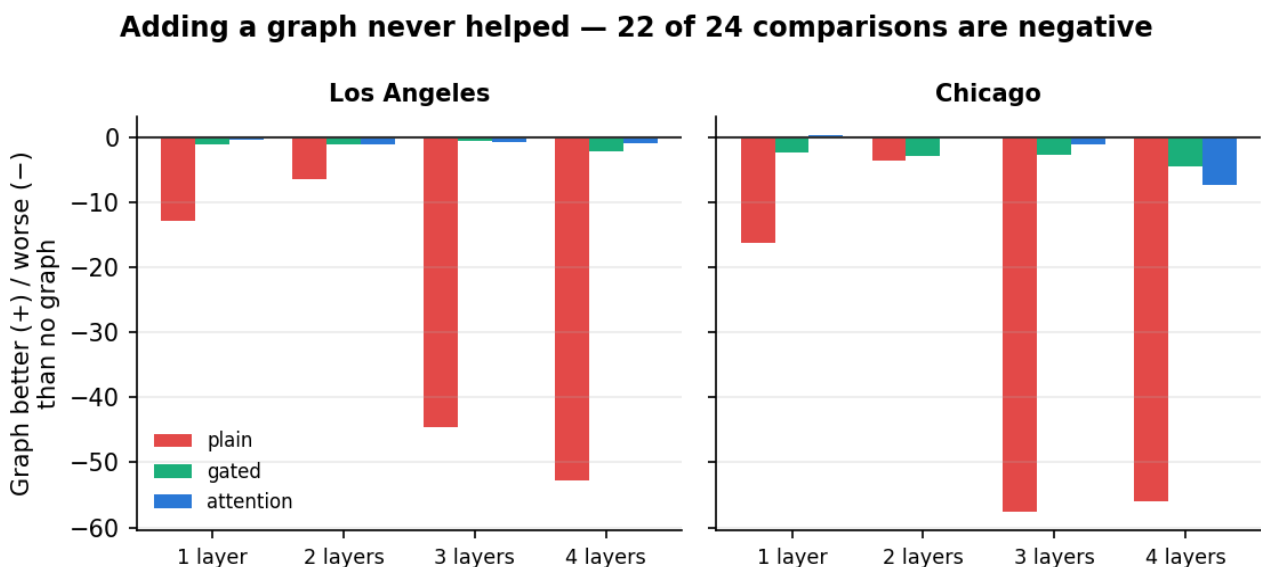
What we tried before finding the answer



Three standard fairness methods, on real Los Angeles data.

We first assumed the gap was real and tried to fix it. Reweighting made it worse. Group-DRO and adaptive thresholds did nothing at any setting. Three reasonable methods failing was the clue that the problem was not what we assumed.

Then we tested graph models thoroughly



3 graph types x 4 depths x 2 cities x 5 runs. Each compared against an equal-sized model with no graph.

Almost every bar points down. Adding a graph did not help — including the gated version. **22 of 24 comparisons were negative.** This matters for the paper: it rules out the explanation that we simply had a weak model.

All results side by side

What we measured	Without graph	With graph	Verdict
Tail F1 (usual fairness score)	23.42	40.38	+16.96 — looks much fairer
Tail AUC (cannot be fooled)	62.03	58.02	−4.01 — actually worse
Tail true-positive rate	23.46	79.91	+56.45 — guessing far more
Tail precision	34.27	27.96	−6.31 — more of them wrong
Head F1 (busy areas)	85.20	84.64	−0.56 — unchanged
Our own base-rate correction	40.97	−2.39	−43.36 — also fooled

The last row is the most important one. We built a correction for the free-points problem, and it reported the gap closing to zero — a perfect result. It was wrong. Correcting for how often crime happens does **not** protect you, because a model can also cheat by changing how often it says "yes". Those are two different problems.

Our main claim: correcting for base rates is not enough. You must also report a score like AUC that cannot be moved by guessing more.

Did it happen twice?

City	Tail F1 change	Statistically real?	Same conclusion?
Chicago	+16.95	yes, $t = 6.49$	yes
Los Angeles	+7.97	yes, $t = 5.65$	yes

Mistakes we found in our own work

These were caught and fixed before writing, not after. Every result above comes from the corrected code.

Problem	How we found it	Fix
Results were not repeatable. The same command gave 48.86 one day and 54.03 the next.	Re-ran an experiment we had already 'finished'.	Set every random seed and forced the GPU to be deterministic. Now identical to 6 decimal places, even after restarting the machine.
We compared a graph model with 4 layers against a no-graph model with only 1 layer.	Checking why a control number had moved.	Every depth now has its own equal-sized control.
A result that looked like a fairness breakthrough.	Checked it with AUC instead of trusting F1.	Withdrawn as a 'win' — it became the paper's main finding instead.

Where the work stands

Item	Status	Note
Evidence and experiments	Done	Reproducible. No further runs needed.
Audit of 7 published papers	Done	New contribution
Effect on decisions	Done	New contribution
'Correction is not enough'	Done	Strongest new claim
fairaudit tool	Done	Supports the paper
Gated GNN as a method	Done	Did not work — negative on both cities
Attack / defence experiments	Excluded	Run with the old faulty code. Suggest a separate paper.
The written paper	Not started	The remaining work
Ethics & limitations section	Not started	Required for predictive policing

Proposed next step

Target **ACM FAccT 2027**. Abstract due **27 October 2026**, full paper **3 November 2026**. Its "Evaluations and evaluation practices" track explicitly invites papers that examine evaluation metrics and measurement practices. Acceptance was 26.8% in 2025, and papers sent to the revision round were accepted 71% of the time.

Question for discussion: the gated-GNN approach came out negative on both cities, so it cannot carry the paper. Should we build the submission around the measurement finding instead, and keep the robustness work for a second paper?